

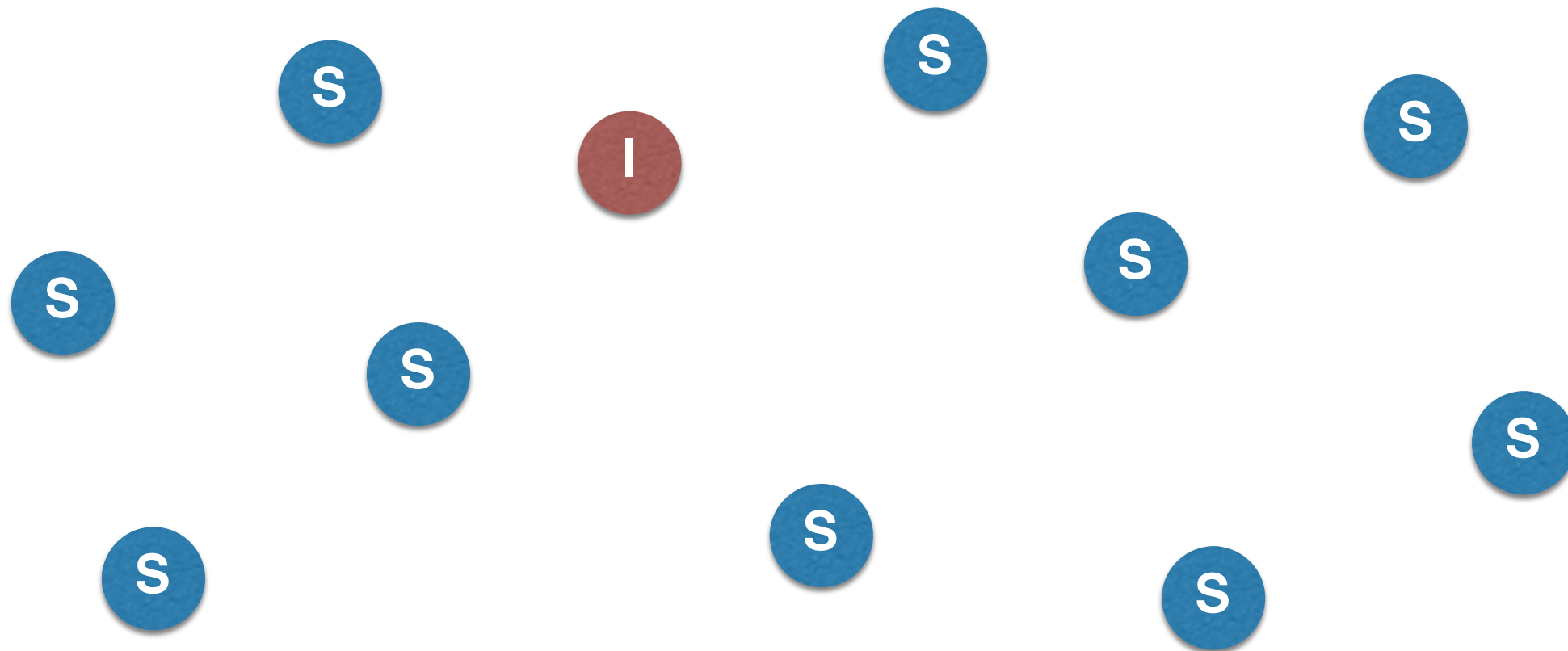
Lecture 4: overview

- **Admin**
- **Continuing modeling activity**
- **More discussion of SIR model of Kermack and McKendrick (1927)**
- **Implications of the SIR model on controlling epidemics**
- **Google trends**

Admin

- **Sign up for two presentation slots by Monday, February 6! If you don't have a team, use Piazza to find one.**
- **Kausalya will schedule practice presentations for the teams presenting “Epidemiology” papers.**
- **What to submit for Lab 0:**
 - Matlab code only as .m files.**
 - In your code, include the comments / answers to lab questions.**
- **Lab next Tuesday in class – bring computer**

General idea

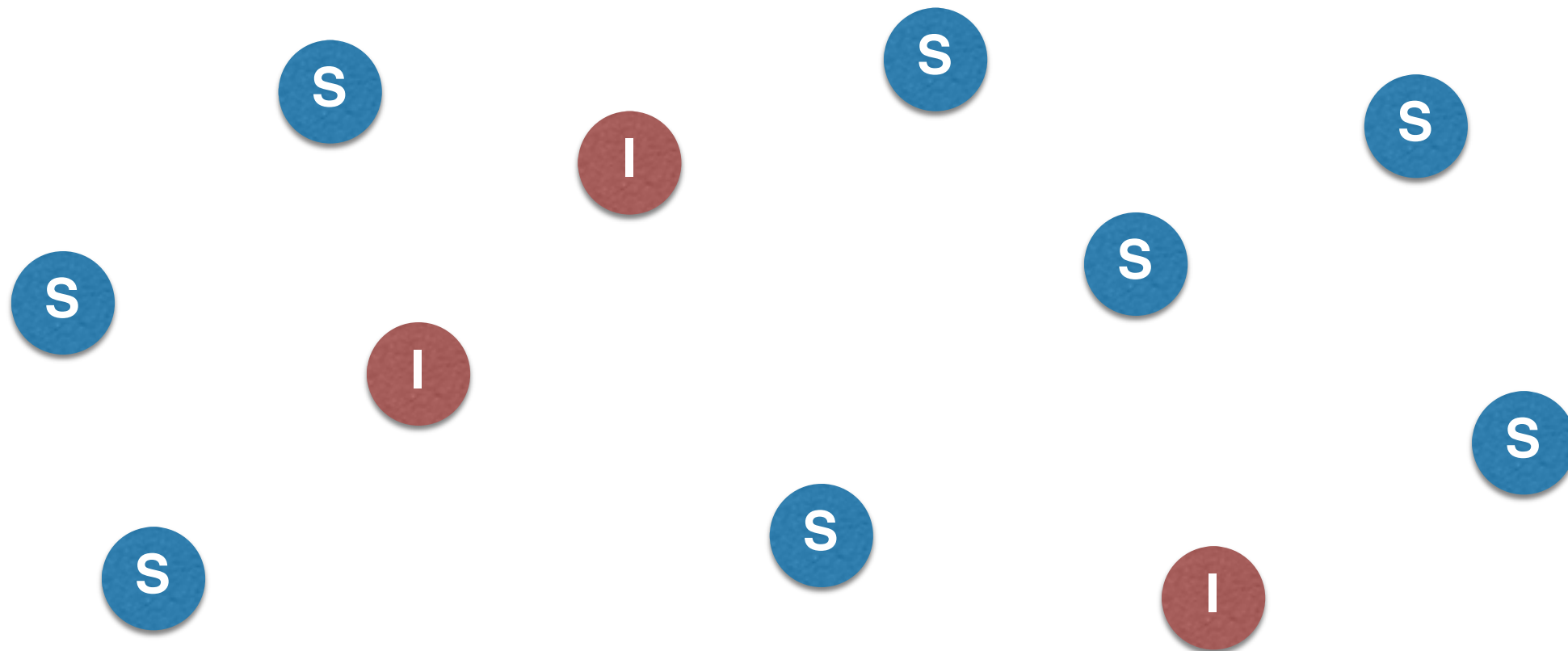


Three populations:

- **Susceptible** $S(t)$
- **Infected** $I(t)$
- **Removed** $R(t)$



General idea

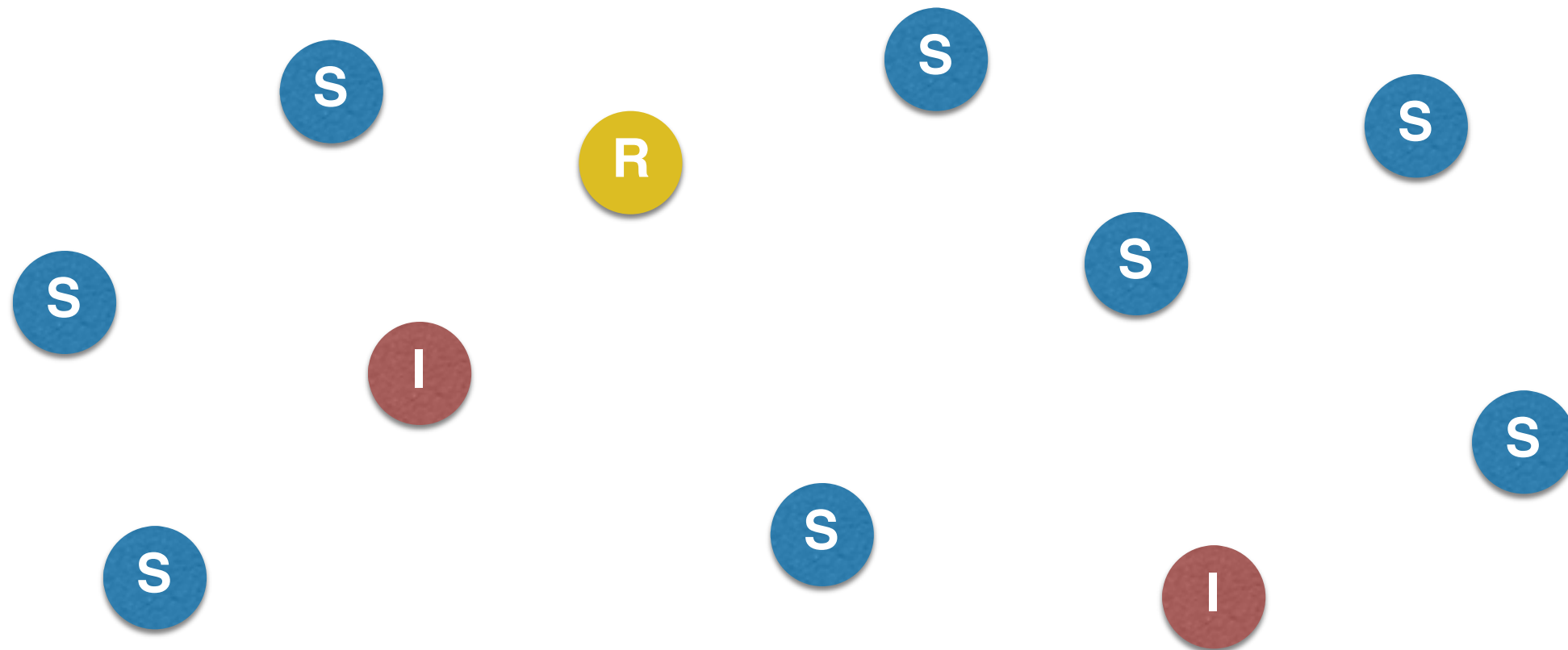


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General idea



Three populations:

- **Susceptible** $S(t)$
- **Infected** $I(t)$
- **Removed** $R(t)$



SIR Model: constant rates

$S(t)$: **Susceptible**

$I(t)$: **Infected**

$R(t)$: **Removed/recovered**

$$\frac{dS}{dt} = -\beta SI$$

Infection rate



$$\frac{dI}{dt} = \beta SI - \gamma I$$

$$\frac{dR}{dt} = \gamma I$$

Recovery/death rate



Modeling activity

Assume total population is constant.

Justifications

- **No births?**
- **Other ideas?**

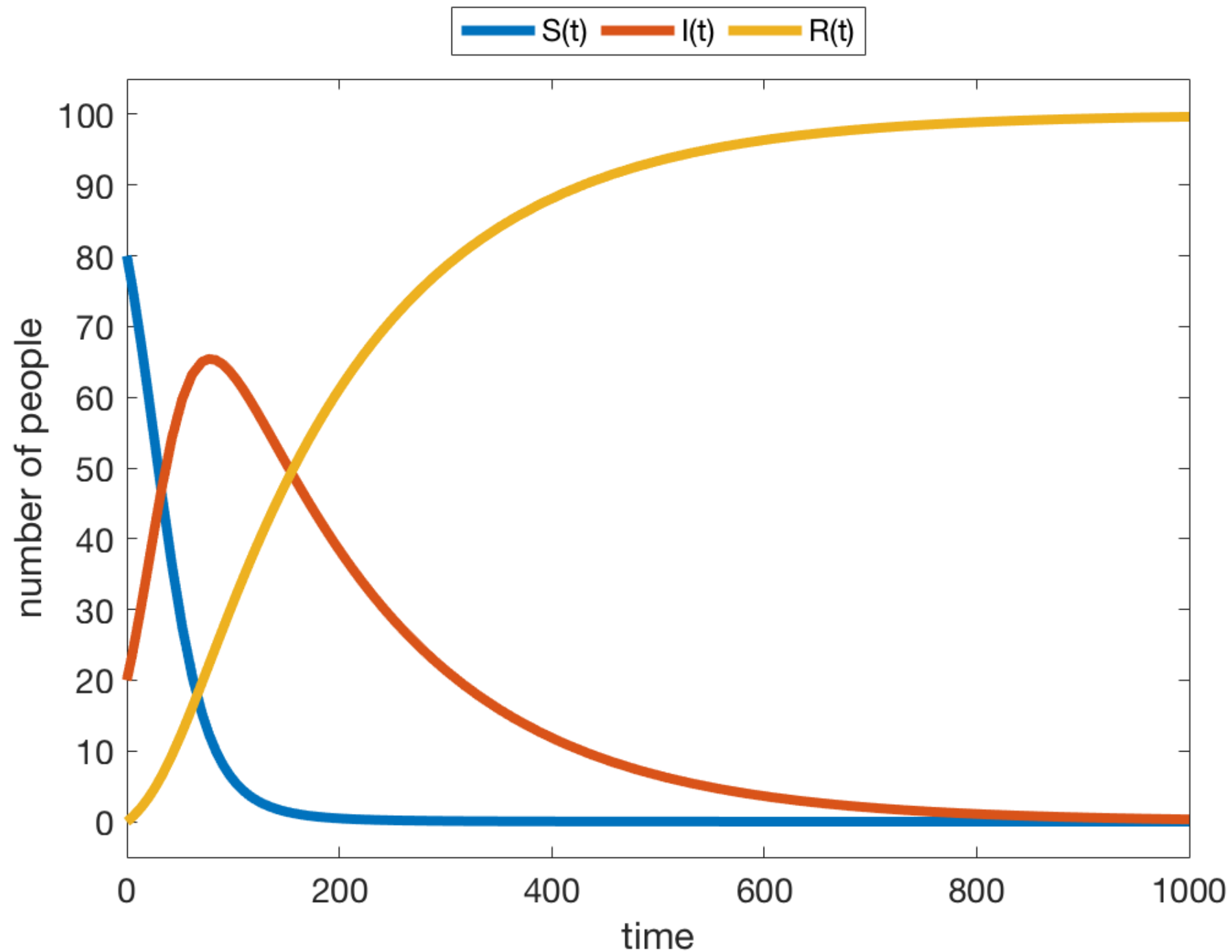
Modeling activity

How do we expect these populations to change in time?

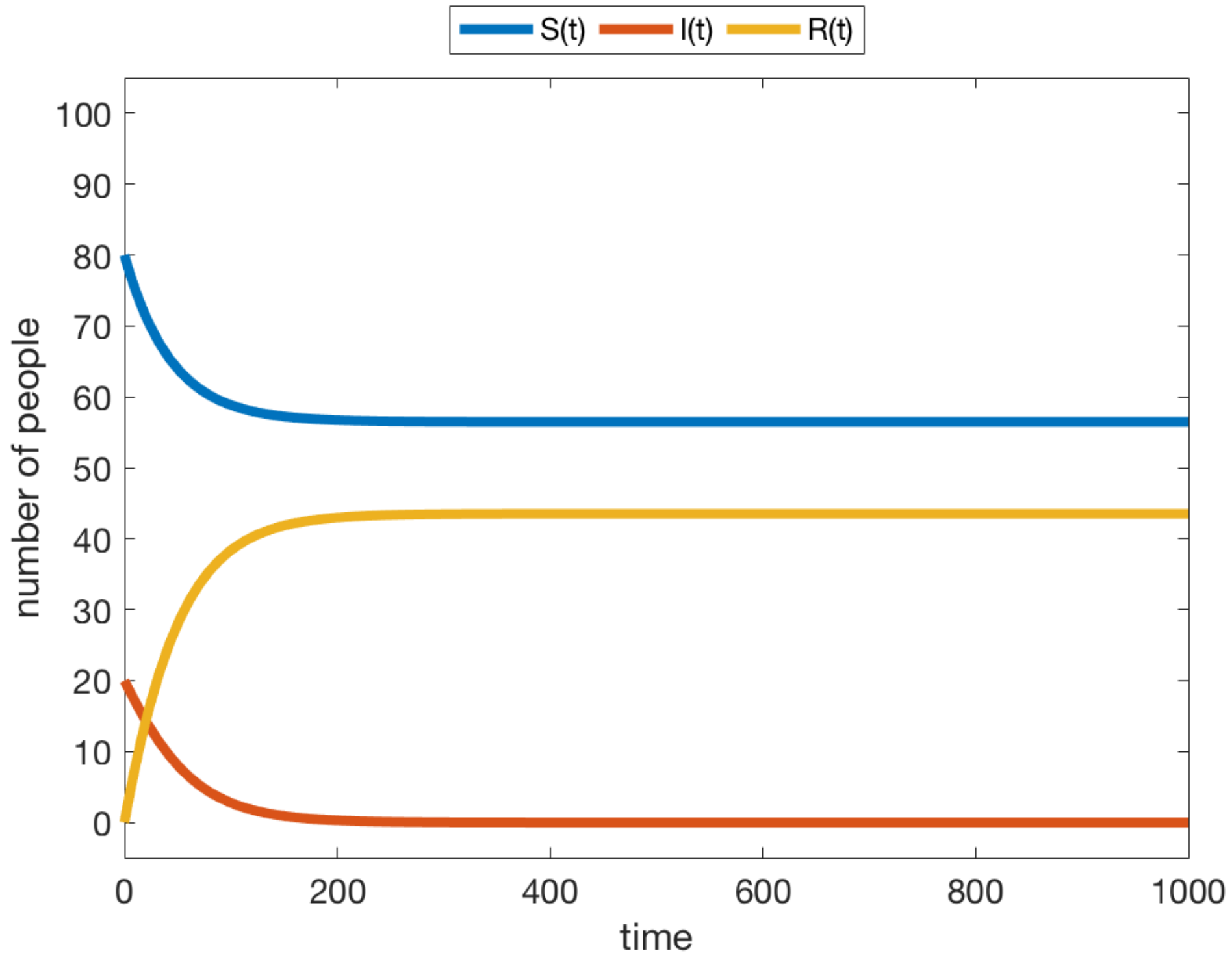
Does it depend on:

- **Initial size of each population?**
- **The infection and recovery rates?**
- **Other ideas?**

Modeling activity



Modeling activity



Continuing the activity

The more complicated model: revisited

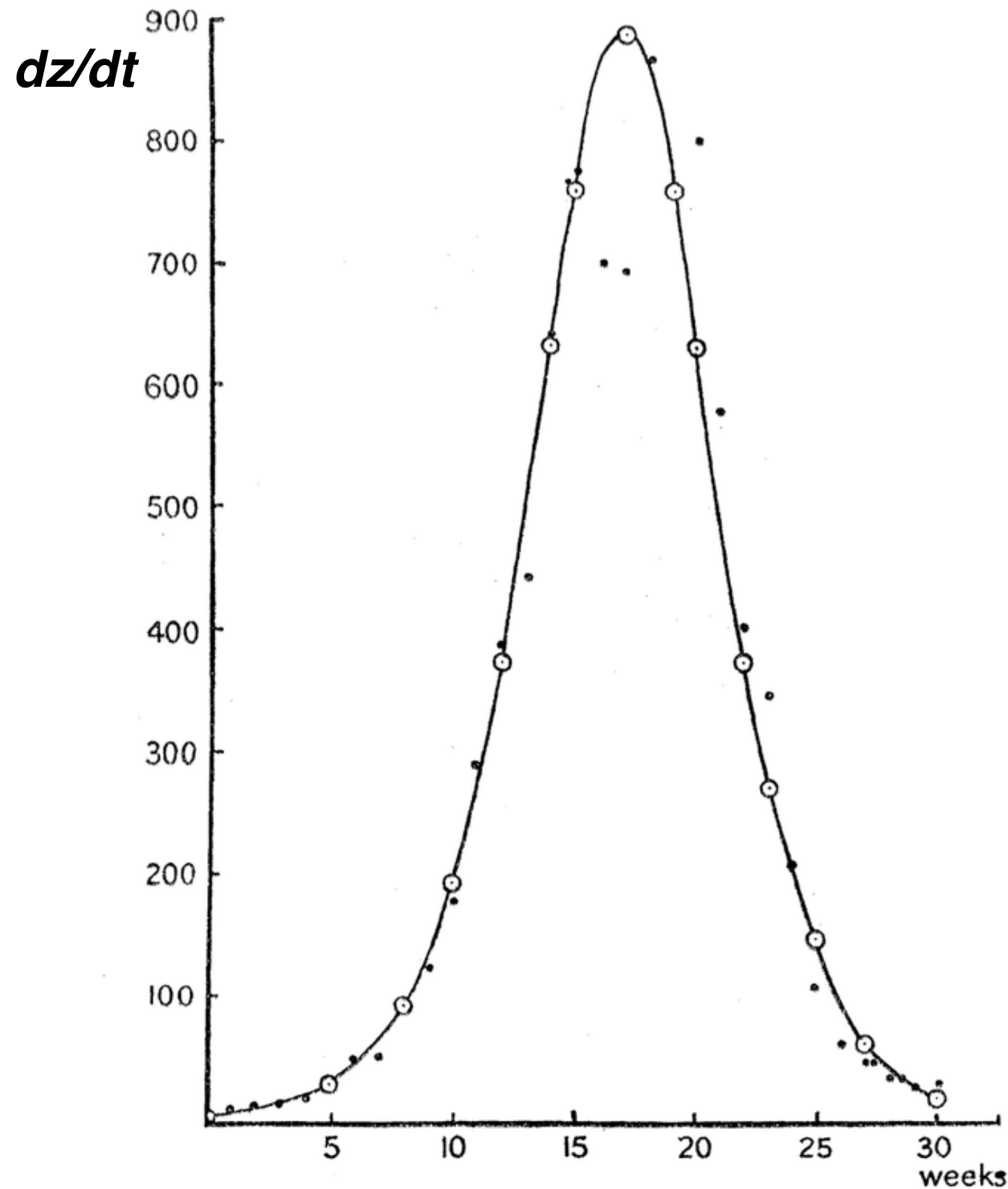
$$x_t + y_t + z_t = N.$$

$$v_t = -\frac{dx_t}{dt},$$

$$\frac{dx_t}{dt} = -x_t \left[\int_0^t A_\theta v_{t-\theta} d\theta + A_t y_0 \right],$$

$$\frac{dz_t}{dt} = \int_0^t C_\theta v_{t-\theta} d\theta + C_t y_0,$$

$$y_t = \int_0^t B_\theta v_{t-\theta} d\theta + B_t y_0,$$

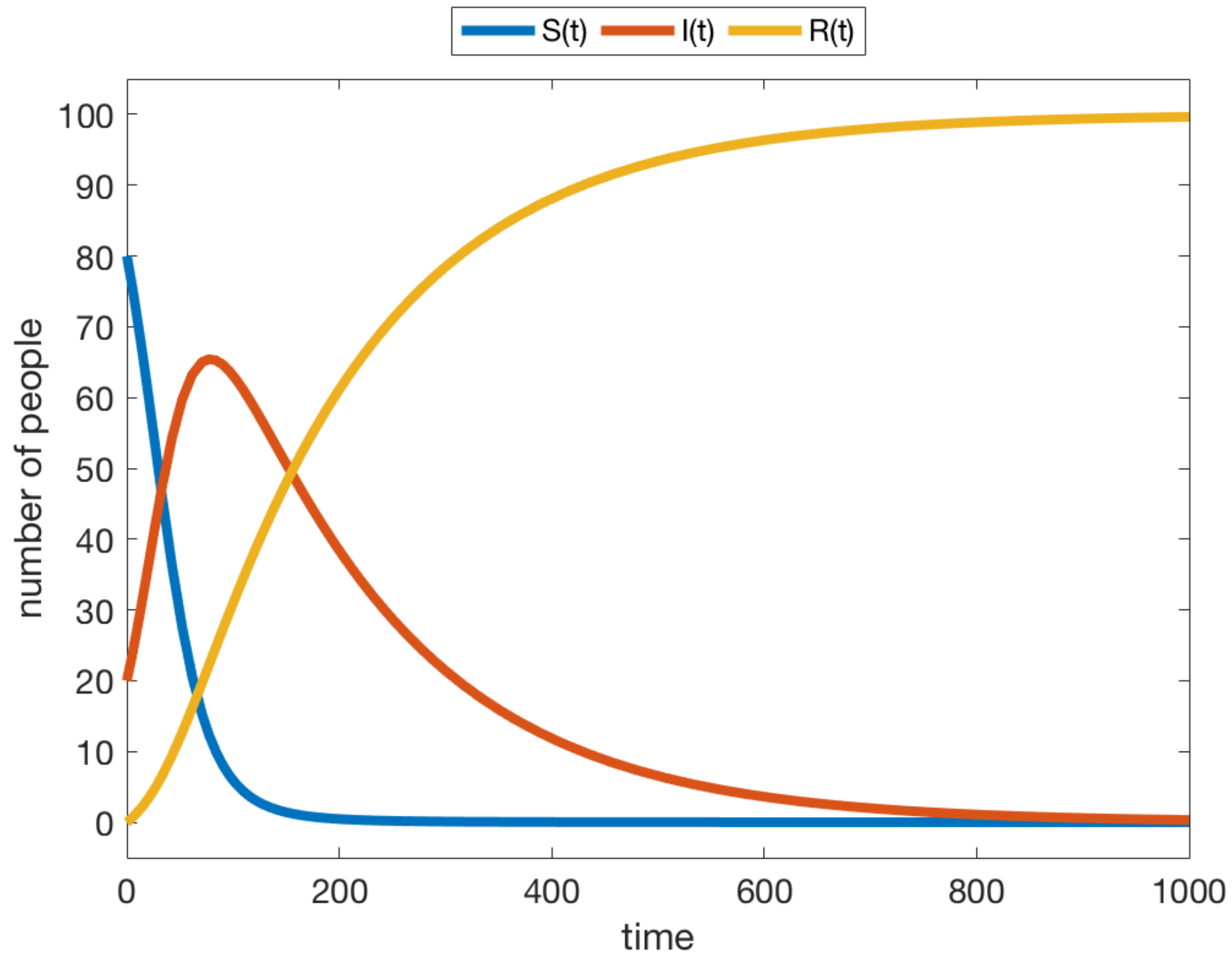


**Is this consistent with
our activity sketches?**

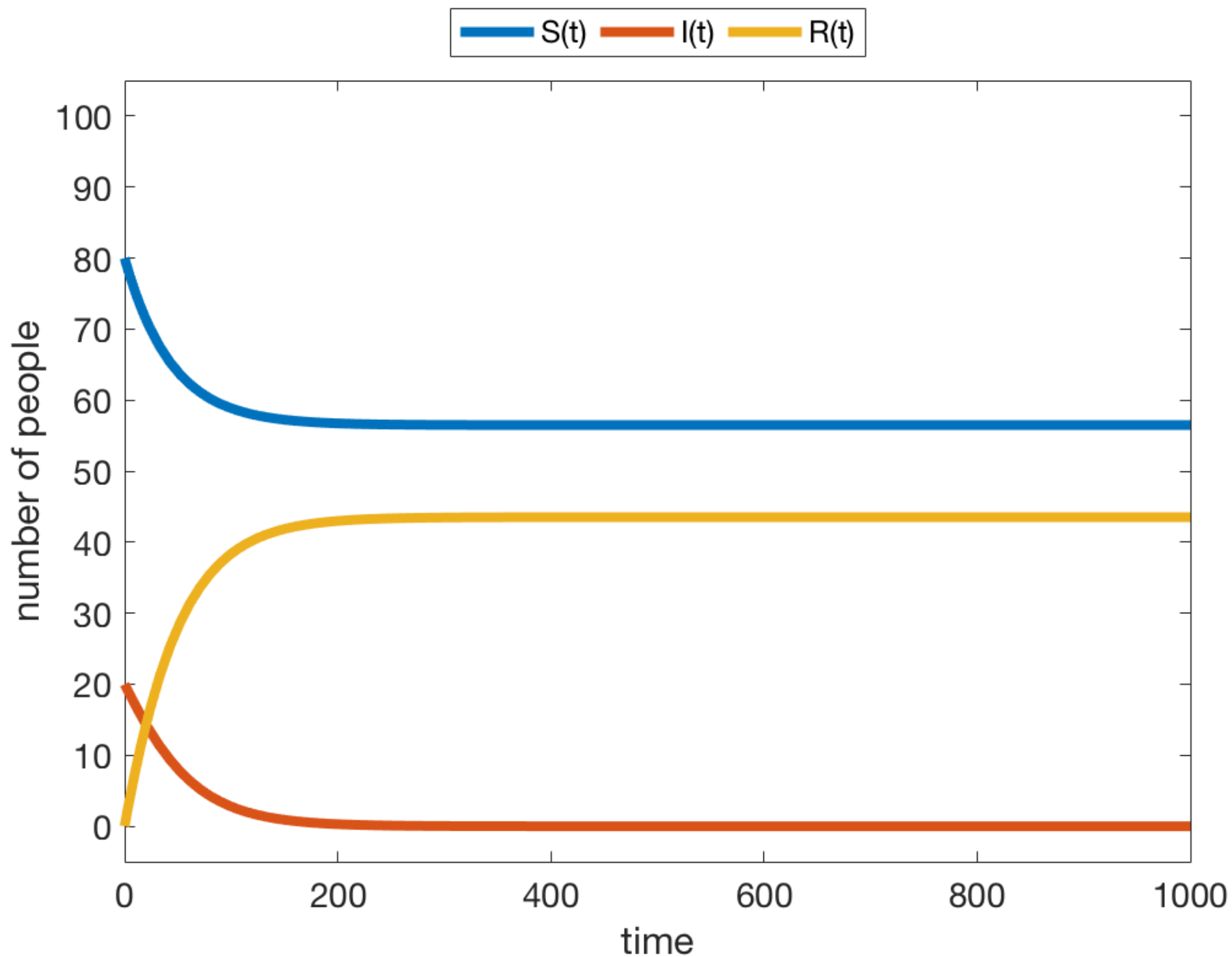
The accompanying chart is based upon figures of deaths from plague in the island of Bombay over the period December 17, 1905, to July 21, 1906. The ordinate represents the number of deaths per week, and the abscissa denotes the time in weeks. As at least 80 to 90 per cent. of the cases reported terminate fatally, the ordinate may be taken as approximately representing dz/dt as a function of t . The calculated curve is drawn from the formula

$$\frac{dz}{dt} = 890 \operatorname{sech}^2(0.2t - 3.4).$$

Bad news case



Good news case



We are, in fact, assuming that plague in man is a reflection of plague in rats, and that with respect to the rat (1) the uninfected population was uniformly susceptible ; (2) that all susceptible rats in the island had an equal chance of being infected ; (3) that the infectivity, recovery, and death rates were of constant value throughout the course of sickness of each rat ; (4) that all cases ended fatally or became immune ; and (5) that the flea population was so large that the condition approximated to one of contact infection. None of these assumptions are strictly fulfilled and consequently the numerical equation can only be a very rough approximation. A close fit is not to be expected, and deductions as to the actual values of the various constants should not be drawn. It may be said, however, that the calculated curve, which implies that the rates did not vary during the period of the epidemic, conforms roughly to the observed figures.

What can we learn?

$S(t)$: **Susceptible**

$I(t)$: **Infected**

$R(t)$: **Removed/recovered**

$$\frac{dS}{dt} = -\beta SI$$

Infection rate

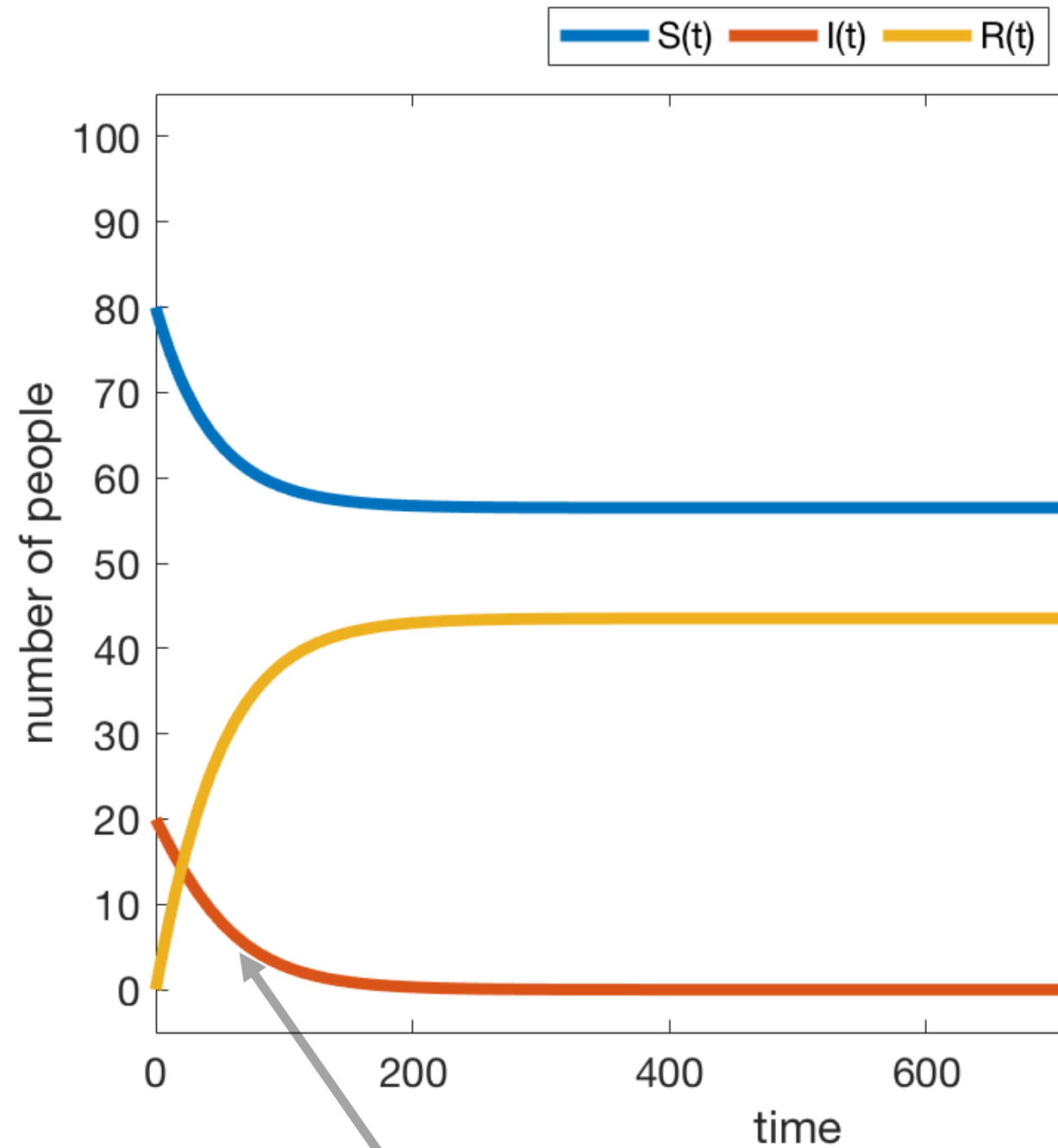
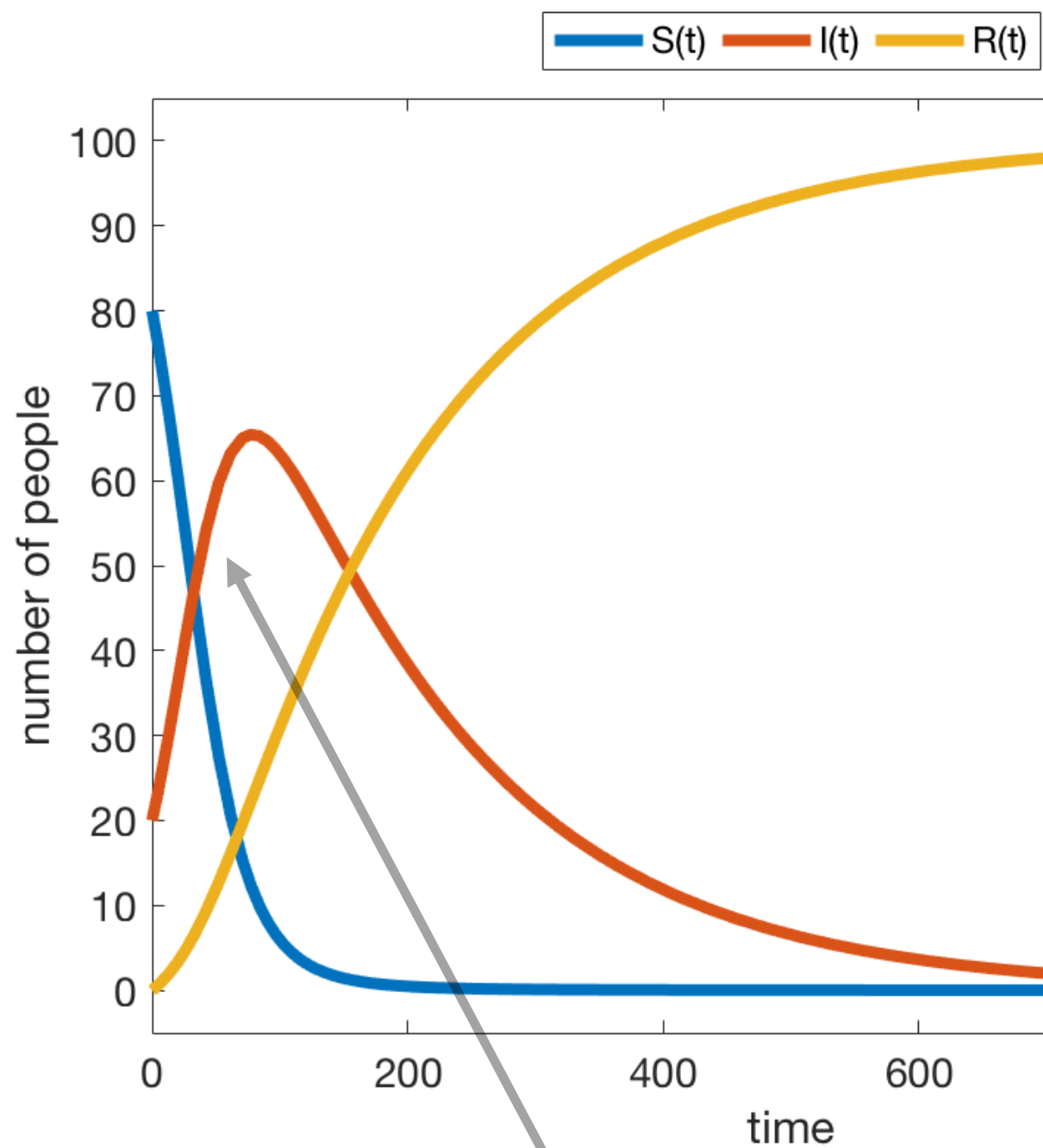


$$\frac{dI}{dt} = \beta SI - \gamma I$$

$$\frac{dR}{dt} = \gamma I$$

Recovery/death rate





Rate of spreading

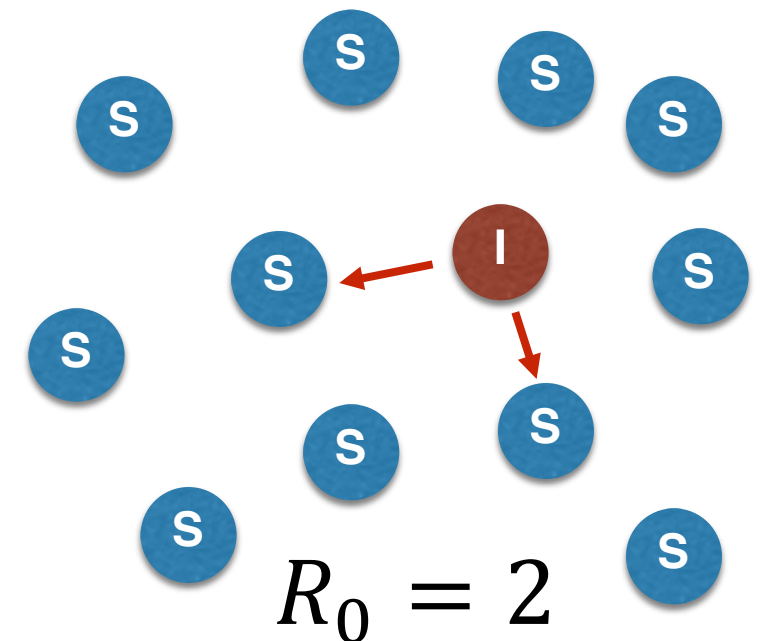
$$\frac{dI}{dt} = (\beta S - \gamma)I > 0$$

β : Infection rate
[units of $(person)^{-1}(time)^{-1}$]

γ : Recovery rate
[units of $(time)^{-1}$]

Infection spreads if $R_0 = \frac{\beta S}{\gamma} > 1$

- R_0 is called the *basic reproduction number*
- It is a *dimensionless* quantity
- It represents the average number of people a single infected individual will infect



Rate of spreading

$$\frac{dI}{dt} = (\beta S - \gamma)I > 0$$

β : Infection rate
[units of $(person)^{-1}(time)^{-1}$]

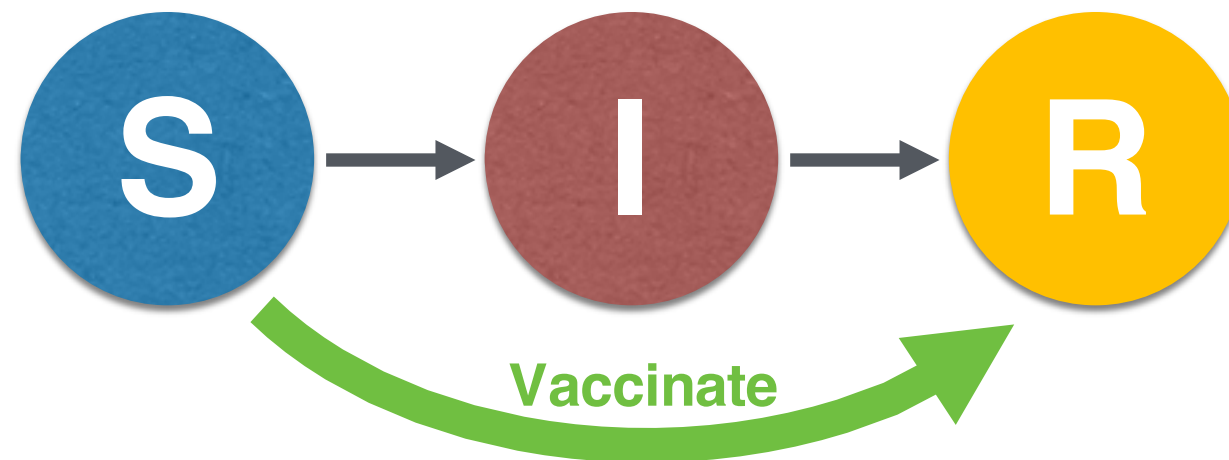
γ : Recovery rate
[units of $(time)^{-1}$]

Infection spreads if $R_0 = \frac{\beta S}{\gamma} > 1$

Techniques to stop the spread of infection:

1. Reduce β (quarantining, restricting movement)
2. Increase γ (treating infected people)
3. Reduce S (vaccinating)

Vaccination



- **Vaccination equivalent to moving a fraction p of the population from S to R**
- **Aim to reduce susceptibles to critical fraction**

$$S_c = \frac{\gamma}{\beta}$$

- **Critical vaccination fraction is therefore**

$$p_c = \frac{\text{Total to vaccinate}}{\text{Total population}} = \frac{S - S_c}{S} = 1 - \frac{S_c}{S} = 1 - \frac{1}{R_0}$$

Critical vaccination fractions

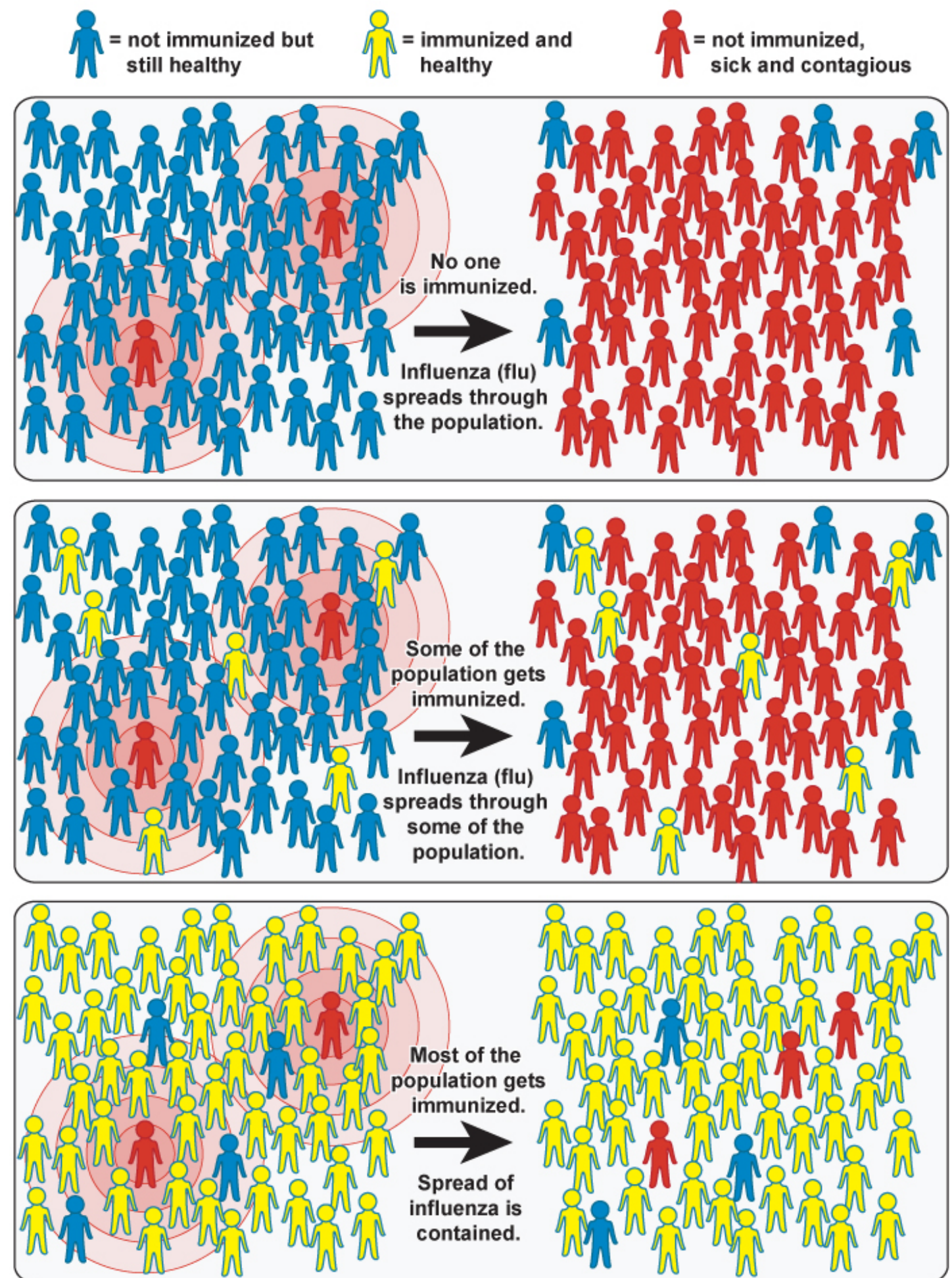
Infectious disease	p_c
Malaria (<i>P. falciparum</i> in hyperendemic regions)	99%
Measles	90–95 %
Pertussis	90–95 %
Human parvovirus infections	90–95 %
Chicken pox	85–90 %
Mumps	85–90 %
Rubella	82–87 %
Polio	82–87 %
Diphtheria	82–87 %
Scarlet fever	82–87 %
Small pox	70–80 %

Source: Anderson & May, Infectious Diseases of Humans, Oxford University Press

“Herd immunity”

- Term used for a population that is sufficiently vaccinated that a disease cannot catastrophically spread
- Possible conflict of interest between an individual and the community

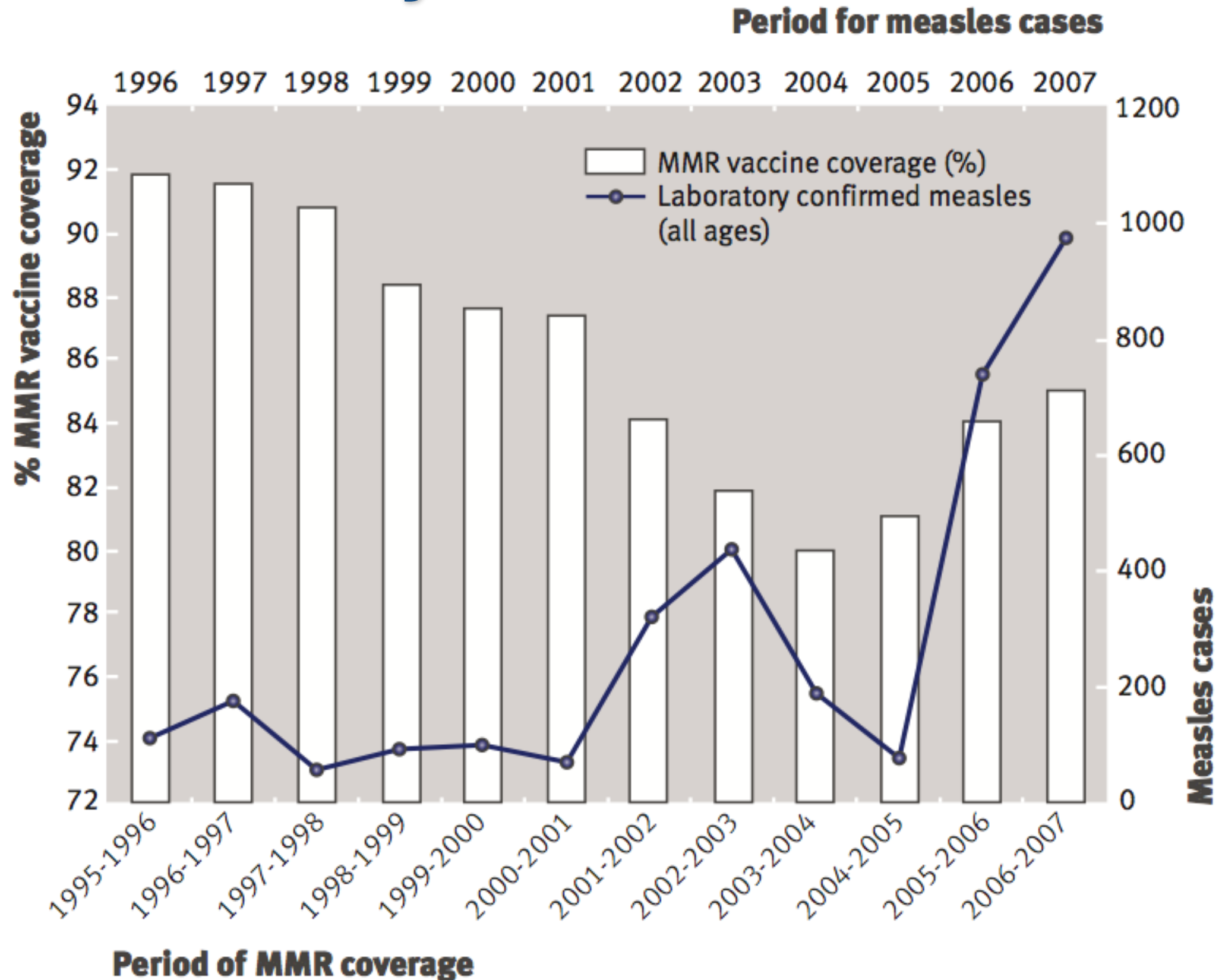
Graphic: Center for Disease Control



Herd immunity failure

- 1998 journal article published in *The Lancet* links MMR vaccine to autism and digestive disease
- Conclusively debunked over the subsequent decade: paper retracted in 2010 and authors tried for professional misconduct
- However, public scare causes many parents to opt out, causing vaccination rates to significantly drop

Herd immunity failure



News article from Feb 4, 2015

U.S. House leaders back inoculations as safety debate rages

BY EMILY STEPHENSON

WASHINGTON | Wed Feb 4, 2015 2:09am IST

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Anthony Fauci, Director of the National Institute of Allergy and Infectious Diseases, testifies about the measles outbreak in the United States before a House Energy and Commerce Oversight and Investigations Subcommittee hearing on Capitol Hill in Washington February 3, 2015.

CREDIT: REUTERS/JIM BOURG

RELATED VIDEO



(Reuters) - Two leaders in the U.S. House of Representatives said on Tuesday all children should be vaccinated, joining a debate that has become a national political flashpoint, as a measles

Conclusions of paper

1. A mathematical investigation has been made of the progress of an epidemic in a homogeneous population. It has been assumed that complete immunity is conferred by a single attack, and that an individual is not infective at the moment at which he receives infection. With these reservations the problem has been investigated in its most general aspects, and the following conclusions have been arrived at.

2. In general a threshold density of population is found to exist, which depends upon the infectivity, recovery and death rates peculiar to the epidemic. No epidemic can occur if the population density is below this threshold value.

3. Small increases of the infectivity rate may lead to large epidemics ; also, if the population density slightly exceeds its threshold value the effect of an epidemic will be to reduce the density as far below the threshold value as initially it was above it.

4. An epidemic, in general, comes to an end, before the susceptible population has been exhausted.

5. Similar results are indicated for the case in which transmission is through an intermediate host.

Google trends and flu data

Google trends

Harvard University Private university in Cambridge, Mas...	Taylor Swift Singer-songwriter	+ Add comparison
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Worldwide ▼

12/30/05 - 1/30/17 ▼

All categories ▼

Web Search ▼

Interest over time ?



<https://www.google.com/trends/>

Correlations

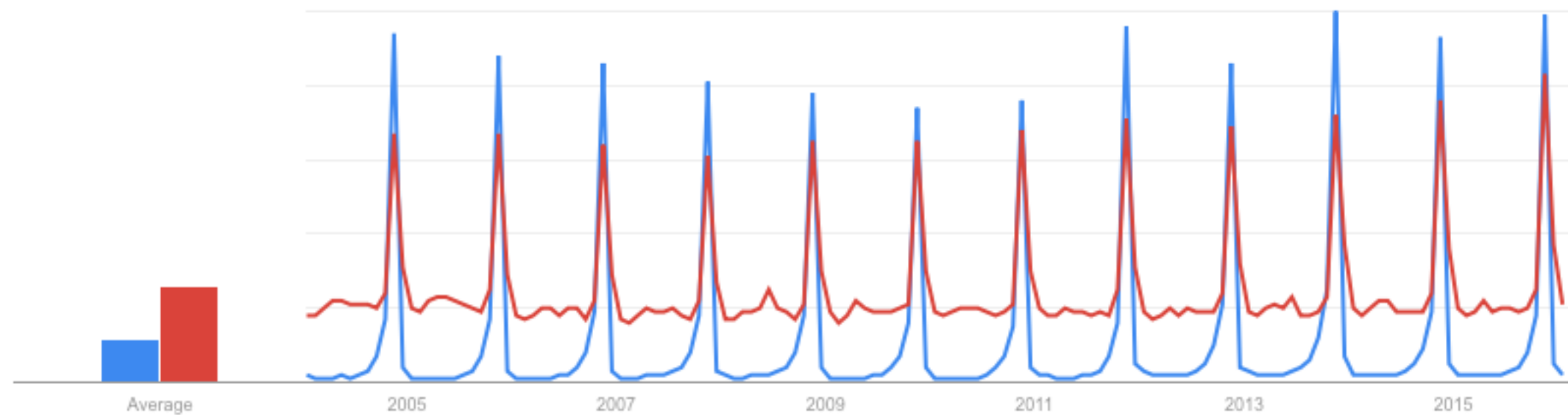
thanksgiving
Search term

turkey
Search term

+ Add term

Interest over time ?

News headlines ? Forecast ?



Correlations

- **Find two search terms that are negatively correlated with each other**
- **Find a search term that has two peaks in a year**
- **Find a search term that has three peaks in a year**

A tool for flu?

● flu
Search term

+ Compare

Worldwide ▼

Past 5 years ▼

All categories ▼

Web Search ▼

Interest over time ?





Detecting influenza epidemics using search engine query data

Jeremy Ginsberg¹, Matthew H. Mohebbi¹, Rajan S. Patel¹, Lynnette Brammer², Mark S. Smolinski¹ & Larry Brilliant¹

¹Google Inc. ²Centers for Disease Control and Prevention

Epidemics of seasonal influenza are a major public health concern, causing tens of millions of respiratory illnesses and 250,000 to 500,000 deaths worldwide each year¹. In addition to seasonal influenza, a new strain of influenza virus against which no prior immunity exists and that demonstrates human-to-human transmission could result in a pandemic with millions of fatalities². Early detection of disease activity, when followed by a rapid response, can reduce the impact of both seasonal and pandemic influenza^{3,4}.

One way to improve early detection is to monitor health-seeking behavior in the form of online web search queries, which are submitted by millions of users around the world each day.

This paper was originally published in *Nature* Vol 457, 19 February 2009,
doi:10.1038/nature07634

<http://dx.doi.org/10.1038/nature07634>

Overview

Traditional surveillance systems, including those employed by the U.S. Centers for Disease Control and Prevention (CDC) and the European Influenza Surveillance Scheme (EISS), rely on both virologic and clinical data, including influenza-like illness (ILI) physician visits. CDC publishes national and regional data from these surveillance systems on a weekly basis, typically with a 1-2 week reporting lag.

In an attempt to provide faster detection, innovative surveillance systems have been created to monitor indirect signals of influenza activity, such as call volume to telephone triage advice lines⁵ and over-the-counter drug sales⁶. About 90 million American adults are believed to search online for information about specific diseases or medical problems each year⁷, making web search queries a uniquely valuable source of information about health trends. Previous attempts at using online activity for influenza surveillance have counted search queries submitted to a Swedish medical website⁸, visitors to certain pages on a U.S. health website⁹, and user clicks on a search keyword advertisement in Canada¹⁰. A set of Yahoo search queries containing the words “flu” or “influenza” were found to correlate with virologic and mortality surveillance data over multiple years¹¹.

Their model

Our proposed system builds on these earlier works by utilizing an automated method of discovering influenza-related search queries. By processing hundreds of billions of individual searches from five years of Google web search logs, our system generates more comprehensive models for use in influenza surveillance, with regional and state-level estimates of influenza-like illness (ILI) activity in the United States. Widespread global usage of online search engines may enable models to eventually be developed in international settings.

By aggregating historical logs of online web search queries submitted between 2003 and 2008, we computed time series of weekly counts for 50 million of the most common search queries in the United States. Separate aggregate weekly counts were kept for every query in each state. No information about the identity of any user was retained. Each time series was normalized by dividing the count for each query in a particular week by the total number of online search queries submitted in that location during the week, resulting in a query fraction (Supplementary Figure 1).

Queries to physician visits

We sought to develop a simple model which estimates the probability that a random physician visit in a particular region is related to an influenza-like illness (ILI); this is equivalent to the percentage of ILI-related physician visits. A single explanatory variable was used: the probability that a random search query submitted from the same region is ILI-related, as determined by an automated method described below. We fit a linear model using the log-odds of an ILI physician visit and the log-odds of an ILI-related search query:

$$\text{logit}(P) = \beta_0 + \beta_1 \times \text{logit}(Q) + \epsilon$$

where P is the percentage of ILI physician visits, Q is the ILI-related query fraction, β_0 is the intercept,

β_1 is the multiplicative coefficient, and ϵ is the error term. $\text{logit}(P)$ is the natural log of $P/(1-P)$.

A power law:

$$\text{Odds of flu-related physician visit} = A(\text{Odds of flu search})^{\beta_1}$$

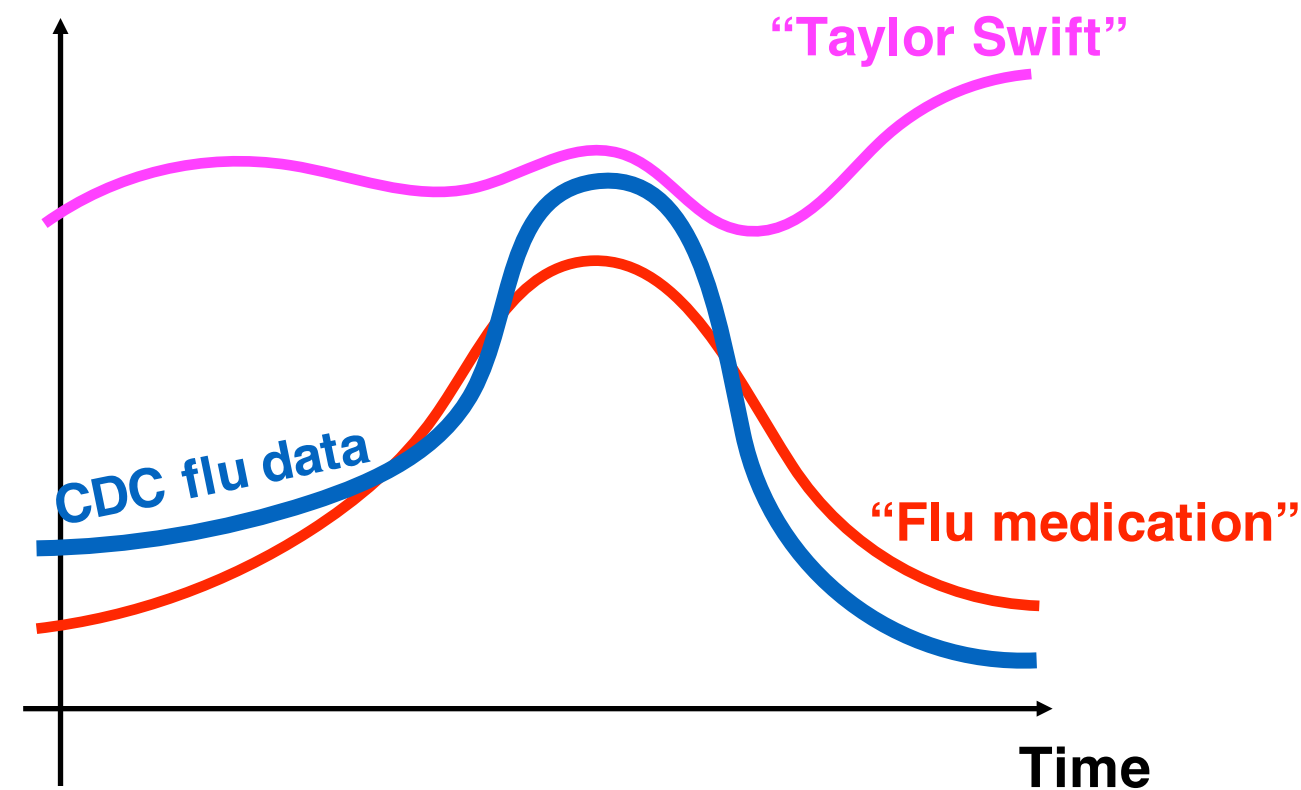
After exponentiation:

$$\frac{P}{1-P} = A \left(\frac{Q}{1-Q} \right)^{\beta_1} e^{\epsilon}$$
$$(A = e^{\beta_0})$$

Obtaining search queries

Publicly available historical data from the CDC's U.S. Influenza Sentinel Provider Surveillance Network¹² was used to help build our models. For each of the nine surveillance regions of the United States, CDC reported the average percentage of all outpatient visits to sentinel providers that were ILI-related on a weekly basis. No data were provided for weeks outside of the annual influenza season, and we excluded such dates from model fitting, though our model was used to generate unvalidated ILI estimates for these weeks.

We designed an automated method of selecting ILI-related search queries, requiring no prior knowledge about influenza. We measured how effectively our model would fit the CDC ILI data in each region if we used only a single query as the explanatory variable Q . Each of the 50 million candidate queries in our database was separately tested in this manner, to identify the search queries which could most accurately model the CDC ILI visit percentage in each region. Our approach rewarded queries which exhibited regional variations similar to the regional variations in CDC ILI data: the chance that a random search query can fit the ILI percentage in all nine regions is considerably less than the chance that a random search query can fit a single location (Supplementary Figure 2).



Finding a set of search terms

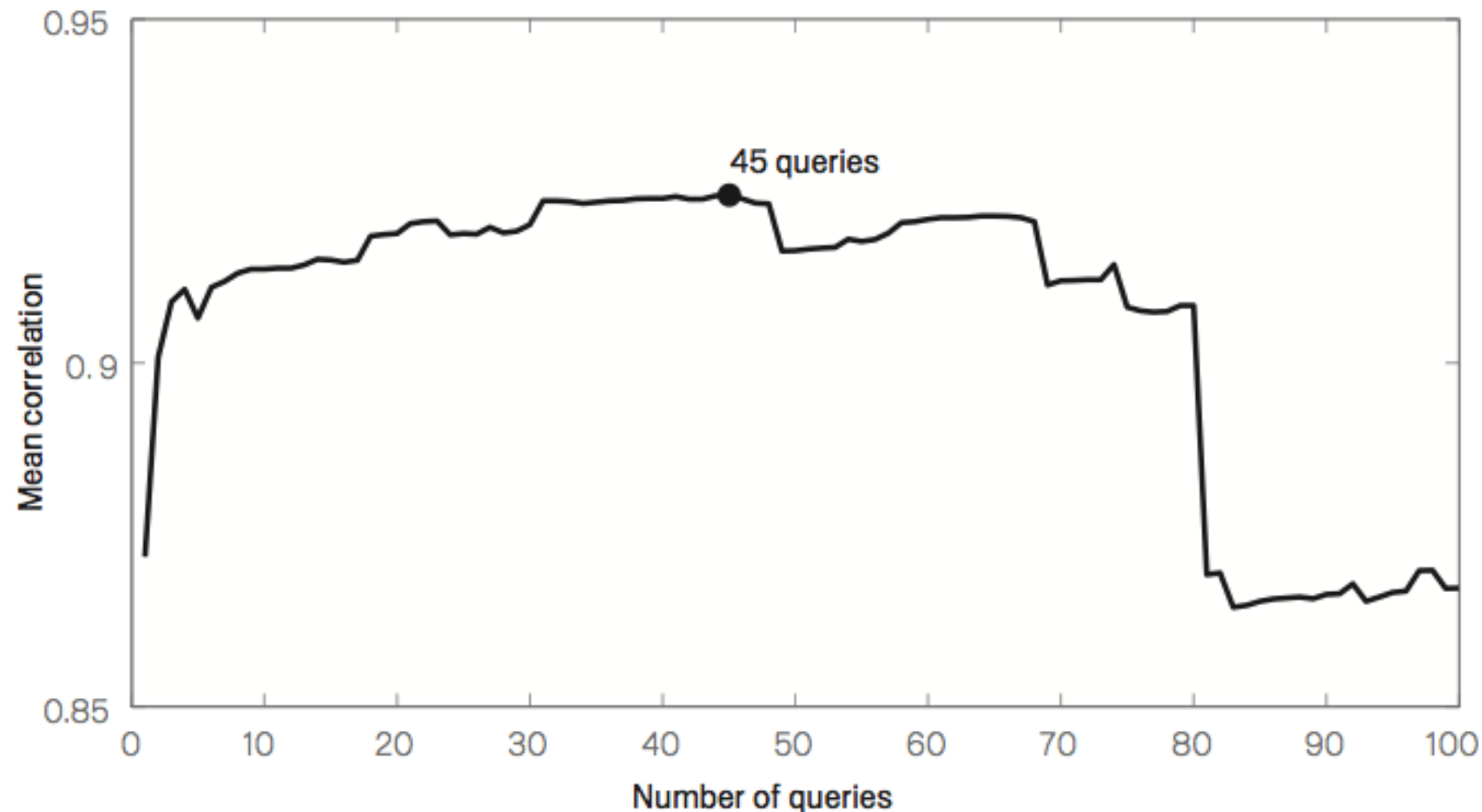


Figure 1: An evaluation of how many top-scoring queries to include in the ILI-related query fraction. Maximal performance at estimating out-of-sample points during cross-validation was obtained by summing the top 45 search queries. A steep drop in model performance occurs after adding query 81, which is “oscar nominations”.

What terms are used

Search Query Topic	Top 45 Queries		Next 55 Queries	
	N	Weighted	N	Weighted
Influenza Complication	11	18.15	5	3.40
Cold/Flu Remedy	8	5.05	6	5.03
General Influenza Symptoms	5	2.60	1	0.07
Term for Influenza	4	3.74	6	0.30
Specific Influenza Symptom	4	2.54	6	3.74
Symptoms of an Influenza Complication	4	2.21	2	0.92
Antibiotic Medication	3	6.23	3	3.17
General Influenza Remedies	2	0.18	1	0.32
Symptoms of a Related Disease	2	1.66	2	0.77
Antiviral Medication	1	0.39	1	0.74
Related Disease	1	6.66	3	3.77
Unrelated to Influenza	0	0.00	19	28.37
	45	49.40	55	50.60

Table 1: Topics found in search queries which were found to be most correlated with CDC ILI data. The top 45 queries were used in our final model; the next 55 queries are presented for comparison purposes. The number of queries in each topic is indicated, as well as query volume-weighted counts, reflecting the relative frequency of queries in each topic.

Validating the model

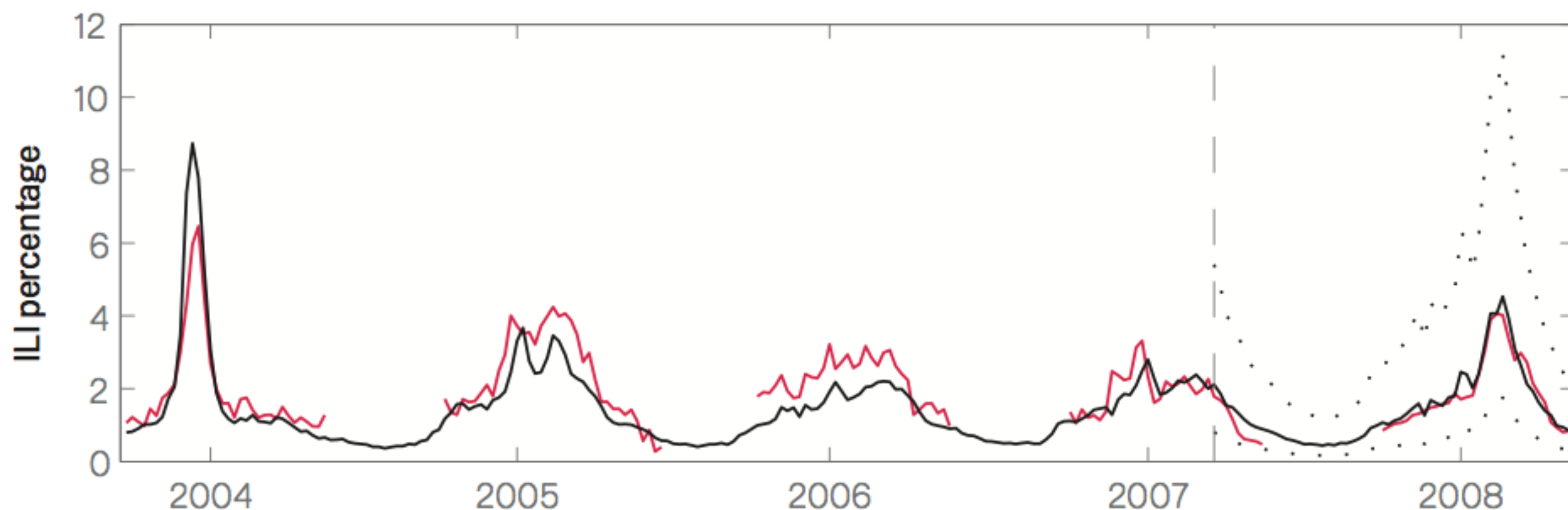


Figure 2: A comparison of model estimates for the Mid-Atlantic Region (black) against CDC-reported ILI percentages (red), including points over which the model was fit and validated. A correlation of 0.85 was obtained over 128 points from this region to which the model was fit, while a correlation of 0.96 was obtained over 42 validation points. 95% prediction intervals are indicated.

Faster than the CDC

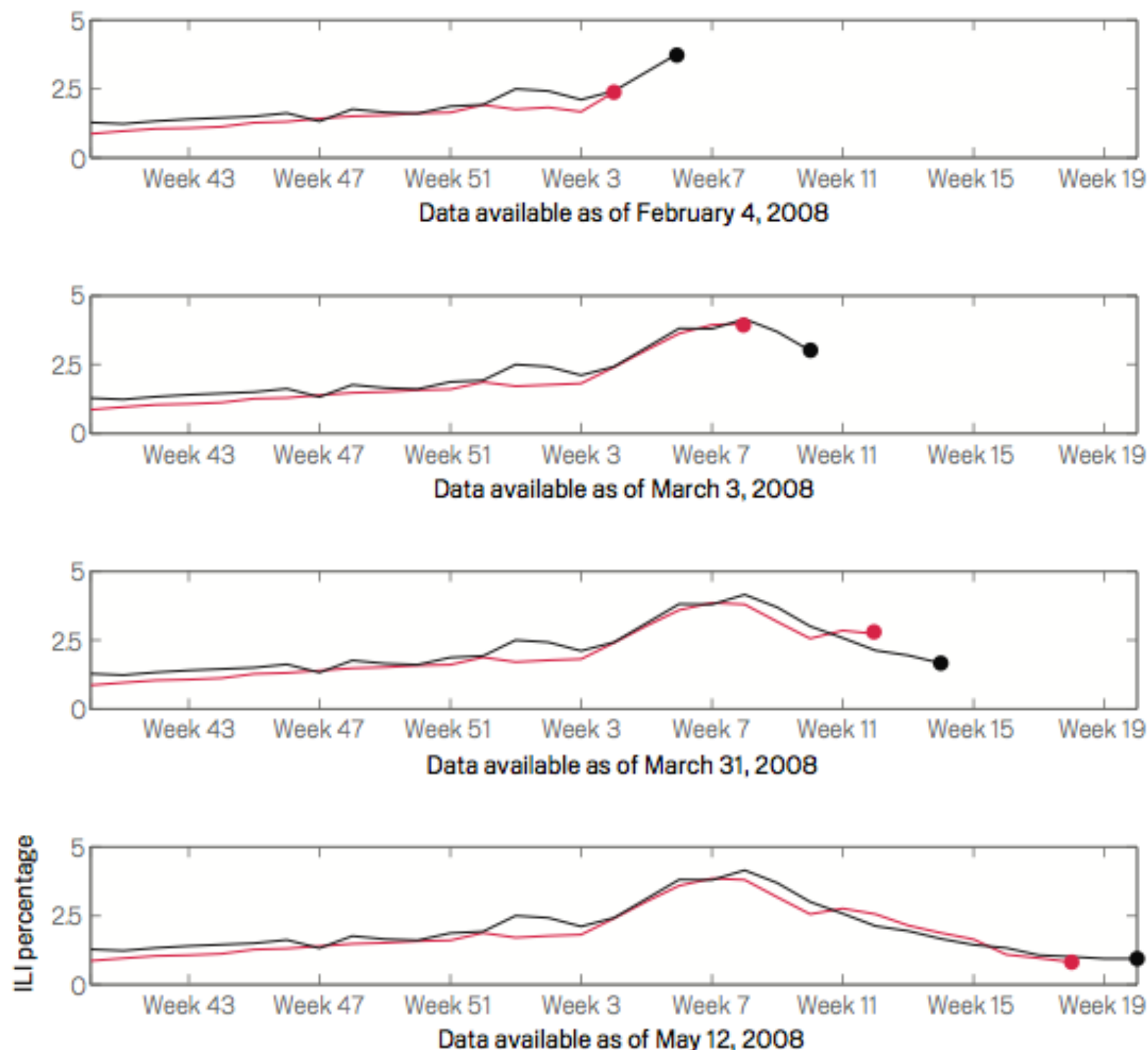


Figure 3: ILI percentages estimated by our model (black) and provided by CDC (red) in the Mid-Atlantic region, showing data available at four points in the 2007-2008 influenza season. During week 5, we detected a sharply increasing ILI percentage in the Mid-Atlantic region; similarly, on March 3, our model indicated that the peak ILI percentage had been reached during week 8, with sharp declines in weeks 9 and 10. Both results were later confirmed by CDC ILI data.

Tuesday lab: SIR model and Google flu data

Bring laptop to class